

Atharva Timsina

Tsukuba, Ibaraki, Japan
070-6578-3785 | timsina@axarva.me | axarva.me | github.com/Axarva

SUMMARY

Software engineer focused on compilers, functional programming, and systems design. Built a statically-typed functional language from first principles in Haskell, contributed to widely-used open-source UI tooling, and developed production backend systems. Strong emphasis on understanding system internals and correctness.

EDUCATION

University of Tsukuba

B.Sc. in Media Sciences and Engineering, School of Informatics

- MEXT (Monbukagakusho) Scholar (Full Government Sponsorship).

Tsukuba, Japan

Apr 2025 – Expected Mar 2029

Osaka University

Center for Japanese Language and Culture (Intensive Program)

- Coursework: Japanese, foundational STEM subjects in Japanese, Japanese Culture.

Osaka, Japan

Apr 2024 – Mar 2025

PROFESSIONAL EXPERIENCE

Colorkrew Inc.

SaaS Software Engineering Intern

- Shipped a priority integration feature that bridges two core company SaaS offerings, resulting in smoother UX.
- Diagnosed and resolved critical N+1 query performance bottlenecks across backend services.
- Owned and executed a comprehensive UI redesign of the platform's primary navigation menu system, improving user discoverability and accessibility.
- Maintained and updated core React components and pushed quick-turnaround hotfixes for styling anomalies across the layout ecosystem.

Tokyo, Japan

Feb 2026 – Present

E. Ruchi Pvt. Ltd.

Backend Engineer

- Architected the backend for a customer loyalty platform serving over 500 concurrent users since its launch in December 2025.
- Designed a decoupled REST API using Express.js and MongoDB, separating survey logic from campaign data.
- Implemented rate-limiting strategies compliant with Brevo API quotas to ensure service stability.
- Developed abuse-prevention systems for reward distribution and profile verification.

Remote / Kathmandu - Contract-based

Oct 2025 – Mar 2026

TECHNICAL PROJECTS

Pura Compiler | Haskell, Megaparsec, Type Theory

Jan 2026

- Designed and implemented a statically-typed functional programming language compiled to JavaScript.
- Engineered a Hindley-Milner Type Inference system to ensure mathematical type safety.
- Implemented a recursive descent parser using Megaparsec, with operator precedence handling and error reporting.
- Implemented an effect-tracking system to enforce explicit I/O permissions at compile time.
- Source code: github.com/Axarva/pura-compiler
- Technical report available at: axarva.me/pura.pdf

XMonad Configuration (Open Source) | Haskell, Linux/X11

- Core contributor to a highly customized tiling window manager configuration with 1,800+ GitHub stars.
- Wrote configurations for multiple open-source projects to create a user interface loved by the Linux community.
- Source code: github.com/Axarva/dotfiles-2.0

LANGUAGES

English (TOEFL iBT score: 111), Nepali (Native), Japanese (Academically Proficient), Hindi (Fluent)

AWARDS

The MEXT (Monbukagakusho) Scholarship: Government of Japan.

National Representative for Nepal: 55th International Chemistry Olympiad (ICHO), ETH Zurich.